

SoundReg: working brief (problem, math, experiment)

1. Problem

From a short multichannel window $\mathcal{A}$ of one ego-mounted microphone array (audio only, no vision at inference), output the set of road users currently making task-relevant sound, each as (class, range, azimuth) in the bird's-eye-view (BEV) plane:

$$\mathcal{S} = \{(c_i, r_i, \theta_i)\}_{i=1}^N, \quad N \text{ unknown and variable.}$$

This is the object list a planner already consumes from LiDAR and camera, not the single direction that classical localization or SELD produce. Sound is useful because it is not line-of-sight bound (sources audible around corners, before they are visible). LiDAR gives accurate positions for in-sight objects and is used as the training label there.

2. Idea (one sentence)

Masking is the acoustic analog of occlusion: a loud source hides weaker concurrent sources in the time-frequency (TF) bins it dominates, so generate objects loudest first (dominance order) and condition each one on those already emitted. The ordering is an inductive bias that should help a finite-capacity model recover low-SNR masked sources; expect the benefit to be largest on hard cases and in the small-data regime, and to shrink with more data.

3. Signal model and the quantities that matter

Stacked-channel STFT of the mixture, with per-source array images $\mathbf{x}_i$ and noise $\mathbf{n}$:

$$\mathbf{X}(t, f) = \sum_{i=1}^N \mathbf{x}_i(t, f) + \mathbf{n}(t, f), \quad E_i = \sum_{t, f} \|\mathbf{x}_i(t, f)\|^2. \quad (1)$$

Difficulty of source j seen against all others (what a parallel detector faces), and after a set C of sources is accounted for:

$$\text{SNR}_j(\emptyset) = \frac{E_j}{\sum_{i \neq j} E_i^{(j)} + E_n^{(j)}}, \quad \text{SNR}_j(C) = \frac{E_j}{\sum_{i \notin C \cup \{j\}} E_i^{(j)} + E_n^{(j)}} \geq \text{SNR}_j(\emptyset), \quad (2)$$

where $E_i^{(j)}, E_n^{(j)}$ are interferer/noise energy in j 's TF support. The useful quantity is the *headroom* $\text{SNR}_j(C) - \text{SNR}_j(\emptyset)$: zero when nothing masks j , large when C holds j 's maskers. $\text{SNR}_j(C)$ is an oracle (perfect removal of C); the model only approaches it by conditioning on emitted tokens. Generation order is set by a dominance score (steered-response power), emitted in descending d_i :

$$d_i = \sum_{t, f} |\mathbf{w}(r_i, \theta_i)^H \mathbf{X}(t, f)|^2. \quad (3)$$

4. Model

Each object is three tokens: class c_i , quantized range t_i^r , quantized azimuth t_i^θ (separate vocabulary per axis; azimuth finer than range). A scene in dominance order π and its likelihood:

$$O_\pi = [\text{BOS}], (c, t^r, t^\theta)_{\pi(1)}, \dots, (c, t^r, t^\theta)_{\pi(N)}, [\text{EOS}], \quad (4)$$

$$p_\Theta(O_\pi \mid \mathcal{A}) = p_\Theta([\text{EOS}] \mid h_{N+1}) \prod_{i=1}^N p_\Theta(c_i \mid h_i) p_\Theta(t_i^r \mid c_i, h_i) p_\Theta(t_i^\theta \mid c_i, t_i^r, h_i), \quad (5)$$

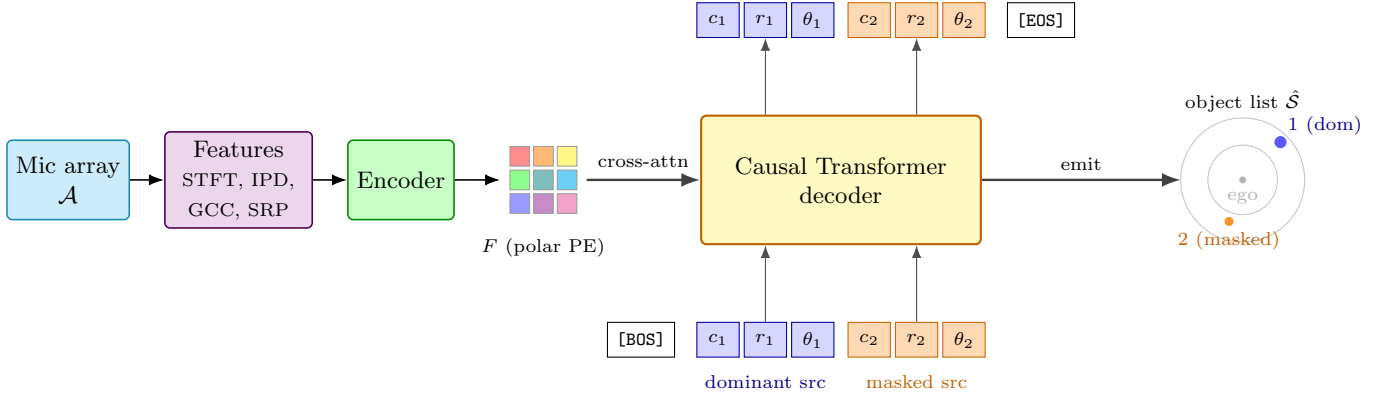


Figure 1: SoundReg. Audio is encoded once into scene features F ; a causal decoder cross-attends to F and emits sounding objects one at a time in dominance order. The dominant source (blue) is emitted first; conditioning on it is intended to let the decoder recover the masked source (orange) that a parallel detector, working from the full mixture, tends to drop.

where h_i summarizes the scene features F and the objects already emitted. Train by teacher forcing with one cross-entropy:

$$\mathcal{L} = - \sum_{i=1}^N \left[\log p_{\Theta}(c_i | h_i) + \log p_{\Theta}(t_i^r | c_i, h_i) + \log p_{\Theta}(t_i^{\theta} | c_i, t_i^r, h_i) \right] - \log p_{\Theta}([\text{EOS}] | h_{N+1}). \quad (6)$$

Pieces to implement:

- Encoder: features $\mathcal{A} \rightarrow$ STFT, inter-channel phase difference (IPD), GCC-PHAT, optional SRP map on the polar grid $\rightarrow F$ with a polar positional encoding (separate range and azimuth embeddings).
- Decoder: causal Transformer, cross-attends to F , token-type masking (class steps emit class tokens, location steps emit location tokens).
- Cardinality: [EOS] ends the sequence (no fixed query count, no threshold).
- Inference: greedy/beam for main numbers; sample S sequences and cluster for existence/uncertainty.

Section 8 documents the architecture as actually built on the Acoustic-BEV data (Fig. 2), the audit of that instantiation, and the proposed revision (Fig. 3).

5. Data and ground truth you need

- Real array+LiDAR driving data (all line-of-sight). Decide and record: array geometry/channels, polar grid $(N_{\theta}, N_r, r_{\max})$, train/val/test split.
- Ground truth = objects *actually emitting* task-relevant sound, with class and position. Sound-activity must be annotated independently of the detector (else the evaluation is circular). Position from the LiDAR track.
- Per-source SNR (for stratifying recall): exact in simulation; on real LOS data estimate by beamforming toward each labeled source.
- NLOS: (i) simulated NLOS scenes with known positions mixed into training of every set-prediction model, (ii) a small hand-annotated real NLOS test set. Scoped demo, not a benchmark.

6. Experiment (run in this order)

Baselines (parallel): GCC/SRP peak picking; SELD-style class+DOA; polar heatmap + peak extraction; query-set predictor (DETR / Sound3DVEDet style). SoundReg is the only sequential one.

Metrics: match prediction to GT by class + polar-distance threshold; report F1, cardinality error $(|\hat{N} - N|)$, range/azimuth MAE. Report mean $\pm$ std over several seeds.

Headline test (do this first). Recall stratified by per-source SNR and by number of concurrent sources, SoundReg vs each baseline.

- Prediction: SoundReg’s recall advantage *grows* as SNR drops and overlap rises.
- Falsifier: a flat advantage. If the gap does not grow at low SNR, the masking idea is wrong, stop here.
- Watch the opposite failure: the decoder can fire [EOS] early and *drop* quiet masked sources, which shows up as *worse* low-SNR recall. Monitor [EOS] calibration; also report a length-controlled decoding variant.

Ablations (each isolates one thing).

- Order: dominance vs distance vs azimuth-sweep vs random vs oracle masking-order (upper bound).
- Conditioning: full vs context-masked decoder (predict each object without seeing previously emitted objects). This is the cleanest test that conditioning, not tokenization, does the work.
- Dominance estimator: beamformer power vs oracle energy vs SNR.
- NLOS: train with vs without simulated NLOS data; compare NLOS detection rate against the query baseline given the *same* NLOS targets.
- Soft target: one-bin Gaussian vs hard label.

Second prediction (data scaling). The advantage should shrink as training data grows (it is an inductive bias, not an information gain). Run a small data-fraction sweep against the query baseline.

7. The data on hand, and what to prepare

Each sample (one token) carries:

```
items[token] = {
  "stft":      ndarray (64, F, T)    # 32 mics; complex split to 64 real ch; encoder input
  "audio":     ndarray (T, 32)       # optional raw multichannel
  "ann":       [ {"center": [x,y,z], "wlh": [w,l,h]}, ... ] # sounding objects only
  "lidar":     ndarray (N, 3)        # box-position source; not used at inference
  "velocity": {lin_x, lin_y, lin_z, ang_x, ang_y, ang_z}    # ego motion (optional)
  "timestamp", "scene_name"
}
```

The annotated boxes are the sounding objects; silent objects are simply not annotated, so there is nothing to detect for them. A single class is used, so the class token c_i in Eqs. (4) to (6) is constant and drops out: a target token is just (r, θ) . Microphone positions come from a separate XML file (the array geometry).

Preparation. Each sample becomes one (features, ordered target sequence) pair:

1. Mic geometry: parse the XML into microphone positions $\in \mathbb{R}^{M \times 3}$. One-time.
2. Polar grid: fix $r_{\max}$, N_r , N_θ . This is the target vocabulary.
3. Steering vectors: precompute $w(r, \theta)$ per grid cell and frequency from the geometry. One-time.
4. Features: from **stft**, build log-magnitude and inter-channel phase difference (optionally GCC-PHAT). Encoder input.
5. Targets: each **ann** center $[x, y]$ gives $r = \sqrt{x^2 + y^2}$, $\theta = \text{atan2}(y, x)$, quantized to a (r, θ) bin.
6. Order: dominance $d_i = \sum_{t,f} |w(r_i, \theta_i)^H \mathbf{X}(t, f)|^2$ per box; sort high to low; prepend [BOS], append [EOS].
7. SNR (evaluation only): beamform toward each source, estimate its SNR, store it with the sample to stratify recall.

After step 7 each sample is $[[\text{BOS}], (r, \theta)_1, \dots, (r, \theta)_N, [\text{EOS}]]$ paired with its features, and the model trains directly on it.

8. Implementation on Acoustic-BEV: as built, audit, revision

As built (v1, Fig. 2). The encoder reuses the MoG-slot network’s front end from **Acoustic-BEV/src/evidential** unchanged: SpeedFiLM conditions the stacked re/im STFT ($64 \times 19 \times 84$) on ego speed, the conv trunk ends in a ($100 \times 19 \times 360$) angular feature map, and the same tokenization the K-slot head used (mean over T , angular axis becomes the sequence) yields 360 memory tokens. Only the head changes: where K learned queries attended once, a causal Transformer decoder cross-attends and emits (t^r, t^θ) pairs in dominance order until [EOS]. Single class, so the class token of Eqs. (4) to (6) drops out. Trained with Eq. (6); scored with the benchmark matcher (5° , $5^\circ + 5$ m gates).

Audit: where v1 under-serves the hypothesis. The red markers in Fig. 2 locate each problem on the component that causes it.

- ① **Time is destroyed before conditioning can use it.** The mean over T integrates out exactly the TF glimpses (frames where the masker is weak) that make a masked source recoverable. The hypothesis needs them downstream.
- ② **Geometry is unused in the features, and range has no axis.** The “angular” axis is a learned 1×1 reinterpretation of channels; nothing ties token k to azimuth k , and range must hide inside channel content. Meanwhile exact steering vectors are already computed, but only for ordering.
- ③ **Vocabulary too coarse for the gate.** 5° azimuth bins against a 5° matching gate donate up to 2.5° of pure quantization error; the continuous-output baseline reports mAAE 2.5° .
- ④ **No operating point.** Baselines tune their threshold for F1 on val; [EOS] gives SoundReg no precision/recall knob, and early [EOS] suppresses recall on precisely the quiet sources under study.
- ⑤ **Architecture reused, weights not.** Training the trunk from scratch forfeits a phase-to-angle mapping that trained MoG checkpoints already contain, on a dataset too small to relearn it comfortably.

Two further, smaller issues do not appear in the figure: dominance is a noisy ordering label near ties (and the free-field steering ignores spatial aliasing above ~ 1.7 kHz and car-body shadowing), and with $N \leq 3$ the loss allocates almost no gradient to the last-emitted quiet objects.

Proposed revision (v2, Fig. 3). Each fix is one green marker, one Settings-level change, one ablation.

- ① Replace the mean with **attention pooling over T** : a learned query per angular position keeps access to per-frame evidence.
- ② Add the **geometric feature path**: steering vectors from the mic XML produce SRP maps on the (r, θ) target grid; a small conv head turns them into polar tokens (range axis restored, azimuth grounded) concatenated into the decoder memory.
- ③ **Azimuth vocabulary** $72 \rightarrow 240$ (1.5° bins); the vocabulary is tiny either way, the quantization floor drops below the gate.
- ④ Report **length-controlled decoding** next to greedy and track [EOS] calibration from epoch 1, so cardinality failures are visible and separable from localization failures.
- ⑤ **Initialize the trunk from a trained MoG checkpoint** (frozen for the first epochs), turning architectural reuse into actual transfer.

The decoder, the dominance ordering, the single cross-entropy, and [EOS] cardinality are unchanged: v2 revises only how evidence reaches the decoder, not the formulation.

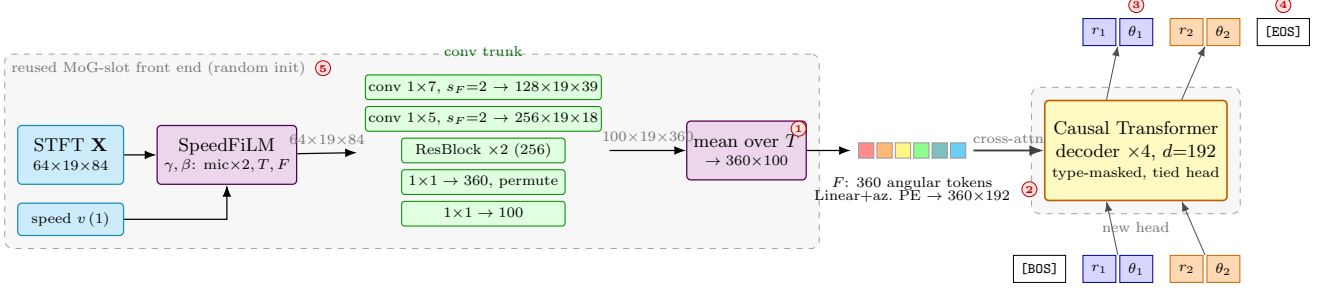


Figure 2: **As built (v1)**. The implemented SoundReg on Acoustic-BEV: the MoG-slot front end is reused unchanged up to its own tokenization (mean over T , 360 angular tokens), and only the head is new. Red markers locate the audit findings of Section 8: ① time pooled out before conditioning, ② learned angular axis with no range axis and no geometric grounding, ③ 5° bins against the 5° gate, ④ no precision/recall knob at [EOS], ⑤ reused architecture but not weights.

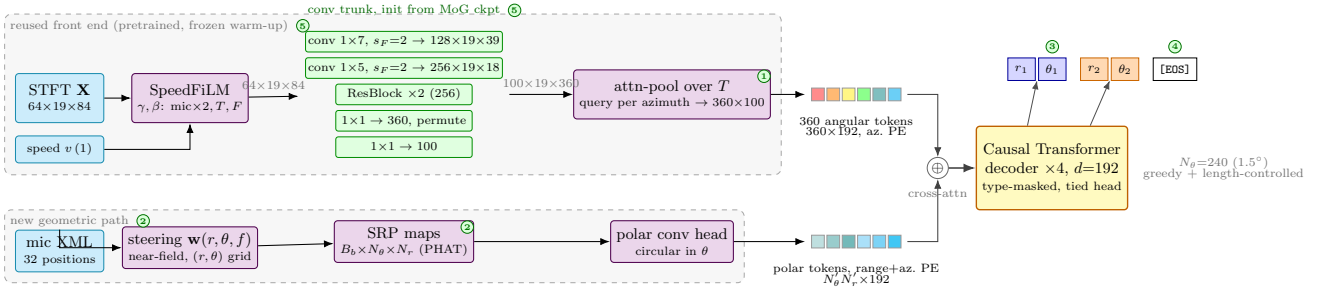


Figure 3: **Proposed revision (v2)**. Same formulation, better evidence flow. Green markers are the fixes of Section 8: ① attention pooling preserves per-frame (masking) structure, ② a geometric path turns the already computed steering vectors into SRP polar tokens with an explicit range axis, ③ 1.5° azimuth bins drop the quantization floor below the gate, ④ length-controlled decoding reported next to greedy with [EOS] calibration tracked, ⑤ the trunk transfers trained MoG weights instead of re-learning phase-to-angle from scratch. Decoder, ordering, loss, and [EOS] cardinality are unchanged.